Lab_Summary

SQL & MQL Functions – Complete Reference

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Oracle SQL Functions

String Functions

Function	Purpose	Example
`lower(string)`	Converts string to lowercase	`select low`
`upper(string)`	Converts string to uppercase	`select upp`
`initcap(string)`	Capitalises first letter of each word	`select ini`
`substr(string, pos, length)`	Returns substring from position	`select sub`
`concat(string1, string2)`	Concatenates two strings	`select con`
`instr(string1, string2)`	Returns position of string2 in string1	`select ins`
`length(string)`	Returns length of string	`select len`
`lpad(string1, length, string2)`	Left-pads string1 with string2 to given length	`select lpa`
`rpad(string1, length, string2)`	Right-pads string1 with string2 to given length	`select rpa`
`ltrim(string)`	Removes leading spaces	`select ltr`
`rtrim(string)`	Removes trailing spaces	`select rtr`
`regexp_like(attr, pattern, opt)`	Matches regular expression (e.g., `i` for case-insensitive)	`regexp_lik`

Numeric Functions

Function	Purpose	Example
`abs(number)`	Absolute value	`select abs(-5) from dual;`
`ceil(number)`	Smallest integer number	`select ceil(3.14) from dual;`
`floor(number)`	Largest integer number	`select floor(3.14) from dual;`
`mod(number1, number2)`	Remainder (modulo)	`select mod(10, 3) from dual;`
`power(number1, number2)`	Raises number1 to number2	`select power(2, 3) from dual;`
`round(number1, number2)`	Rounds to number2 decimal places	`select round(3.14159, 2) from dua`
`trunc(number1, number2)`	Truncates to number2 decimal places	`select trunc(3.14159, 2) from dua`

Date Functions

Function	Purpose	Example
`add_months(date, number)`	Adds months to date	`select add_months(sysdate, 3)`
`next_day(date, weekday)`	Returns next occurrence of weekday	`select next_day(sysdate, 'MON`
`last_day(date)`	Returns last day of the month	`select last_day(sysdate) from`
`current_date` / `sysdate`	Returns current date/time	`select current_date from dual`
`months_between(date1, date2)`	Months between two dates	`select months_between(sysdate`
`new_time(date, zone1, zone2)`	Converts time zone	`select new_time(sysdate, 'EST`
`to_date(string, format)`	Converts string to date	`select to_date('20 MAR 2026',`
`to_char(date, format_mask)`	Converts date to formatted string	`select to_char(sysdate, 'yyyy`

Conversion Functions

Function	Purpose	Example
`to_char(date, format_mask)`	Date to string	`select to_char(sysdate, 'MM/DD/`
`to_date(string, date_format)`	String to date	`select to_date('20 MAR 2026', '`
`to_number(string, format)`	String to number	`select to_number('123.45', '999`
`coalesce(arg1, arg2, …)`	First non-null argument	`select coalesce(cga, 0) from St`
`nvl(arg1, arg2)`	Returns arg2 if arg1 is null	`select nvl(cga, 0) from Student`
`decode(expr, search1, result1, …)`	Case-like conditional	`select decode(unitId,'COMP','CS`

Aggregate Functions

All aggregate functions ignore `NULL` except `count(*)`.

Function	Purpose	Example
`avg(attr)`	Average of values	`select avg(cga) from Student;`
`count(attr)`	Number of non-null values	`select count(studentId) from Student;`
`max(attr)`	Maximum value	`select max(cga) from Student;`
`min(attr)`	Minimum value	`select min(cga) from Student;`
`stddev(attr)`	Sample standard deviation	`select stddev(cga) from Student;`
`sum(attr)`	Sum of values	`select sum(cga) from Student;`

Window Functions

Aggregate Window Functions

Function	Purpose
`avg(expression) OVER (window)`	Average per partition/window
`count(*) OVER (window)`	Count per partition
`max(expression) OVER (window)`	Maximum per partition
`min(expression) OVER (window)`	Minimum per partition
`sum(expression) OVER (window)`	Sum per partition

Value Functions

Function	Purpose
`first_value(expression) OVER (window)`	Value of first row in window frame
`last_value(expression) OVER (window)`	Value of last row in window frame
`lag(expression, offset, default) OVER (window)`	Row n rows before current
`lead(expression, offset, default) OVER (window)`	Row n rows after current
`nth_value(expression, n) OVER (window)`	Value of nth row in window frame

Ranking Functions

Function	Purpose
`row_number() OVER (window)`	Sequential row number (1-based)
`rank() OVER (window)`	Rank with gaps for ties
`dense_rank() OVER (window)`	Rank without gaps
`ntile(n) OVER (window)`	Divides into n buckets
`percent_rank() OVER (window)`	Percentile rank in [0,1]
`cume_dist() OVER (window)`	Cumulative distribution

Window Clause Syntax

```
<window_function> OVER (
    PARTITION BY column1, column2, ...
    ORDER BY column1 [ASC|DESC], ...
)
```